



Open Data, Design, & Development at the Office of Natural Resources Revenue

Beyond open data

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Background

At the Office of Natural Resources Revenue (ONRR), we were already well positioned to comply with new government open-data mandates that have been signed into law over the past year. We have a long history of publishing our federal natural resources revenue data – first in books, then on CDs, and then in an online format in the early 2000s. We took it a step further after joining the Extractive Industries Transparency Initiative (EITI) in 2011, which requires transparency and accountability in managing oil, gas, and mineral resources. Our implementation of the EITI standard required us to not only disclose revenue from energy and mineral extraction, but also required us to show how revenue moves through the government and benefits the public.

We published the first natural resources revenue website in 2015. Although the US withdrew from EITI in 2017, we committed to aligning with the current administration's Management Agenda and providing open data to support good governance and transparency. We continue to embrace the principles of transparency and continue to work in the open, with open source code, and design with user needs in mind.

As a unique steward of U.S. natural resource revenue data, we are informing policy debates and raising public awareness by building the definitive source of timely and useful data about how the government manages federal energy and mineral resources, revenue, and disbursements.

Because we are internally driven to maintain high standards, we stay current with international trends in open data and are constantly searching for ways for our data to go beyond what is expected. We are happy to report that we have already implemented many of the guidelines mandated by recently passed federal laws.

21st Century Integrated Digital Experience Act (IDEA)

The [21st Century Integrated Digital Experience Act \(IDEA\)](#) was signed into law in December 2018.

Here are some of the requirements of 21st Century IDEA:

Section 508 compliance (accessibility)

Our content strategist is well versed in accessibility compliance and works with our team to make sure our content is accessible. He documented simple procedures in the [accessibility section](#) of our [content guide](#) to help team members create content on their own. Our content strategist then performs quarterly accessibility audits, and we set out to fix the errors. In addition to the work we are doing on our own website, our content strategist is creating a workshop to share how to comply with accessibility requirements with other groups within ONRR.

Avoid duplication

This provision states that a website “does not overlap with or duplicate any legacy websites, and, if applicable, ensures legacy websites are reviewed,

eliminated, and consolidated.” [Natural Resources Revenue Data](#) has fully replaced both the EITI website and a public statistics website by combining the data from both websites in one location.

Searchable

We already have a search function on our website.

User-driven design

This provision states that the website must be “designed around user needs with data-driven analysis influencing management and development decisions, using qualitative and quantitative data to determine user goals, needs, and behaviors” and that we should “continually test the website...to ensure that user needs are addressed.” We follow a user-centered development practice to design our site led by our user experience expert. As such, our user experience designer guides our [user-centered design and development cycle](#).

Mobile Devices

We already design for and test our website’s functionality across different browsers and mobile devices to ensure proper functioning.

The Open, Public, Electronic, and Necessary (OPEN) Government Data Act

The Open, Public, Electronic, and Necessary (OPEN) Government Data Act was signed into law in January 2019. The [OPEN Government Data Act](#) was included in the Foundations for Evidence-Based Policymaking Act (Public Law 115-435) as Title II. The OPEN act is more extensive than the 21st Century IDEA. The law expands upon important features of a 2013 Executive Order on Open Data. We already adhere to many of the OPEN act’s key directives.

Here are some of the requirements of the OPEN Government Data Act:

Open data

Our website is built and maintained by [open source code](#) for the purpose of providing [open data](#). We publish as much natural resources revenue data as possible, without revealing proprietary company information or personally identifiable information.

Machine-readable

Agencies need to ensure that any public data asset of the agency is machine-readable. We refrain from creating content that isn't machine readable by default, such as PDFs, and are working to convert legacy content into machine readable formats. We also publish data in both Excel and CSV formats, so viewing and using the data isn't dependent on specific software applications.

Outside collaboration

Agencies must facilitate collaboration with non-Government entities (including businesses), researchers, and the public for the purpose of understanding how users value and use government data. In addition to extensive user testing, we also welcome collaboration on our source code and data through GitHub and invite users to email us with questions and suggestions. Our agile content management, design, and development practices enable our team to respond quickly to user needs, as we did recently when a user contacted us about ambiguous content.

User-centered design/outreach

Agencies must engage the public in using public data assets of the agency and encourage collaboration. In addition to being able to collaborate via [GitHub](#), we also maintain an email address and social media accounts as avenues for public interaction. We conduct extensive user testing as we iterate our

product and include this feedback in our final product. We also participate in outreach via public forums such as conferences and workshops.

Analytics

Agencies must identify and implement methods for collecting and analyzing digital information on data asset usage by users within and outside of the agency, including designating a point of contact within the agency to assist the public and to respond to quality issues, usability issues, recommendations for improvements, and complaints about adherence to open data requirements within a reasonable period of time. While we won't lead the agency initiative on this directive, we already maintain monthly [analytics](#) to help us identify and cater to our audience. Our users can also always reach us via [GitHub](#), social media, or our email address: nrrd@onrr.gov.

Metrics

Agencies must develop and implement a process to evaluate and improve the timeliness, completeness, consistency, accuracy, usefulness, and availability of open Government data assets. We have instituted [metrics](#) that address the above requirements and track them on a regular basis.

Conclusion

Given that we are continually assessing our performance using internal metrics that encompass the areas addressed by IDEA and OPEN Government Data Act, we are confident that we will continue to meet or exceed federal laws and regulations governing open data. We also expect these metrics to evolve as we identify new opportunities through our mission to set the gold standard for open government data.

Note : Reference in this blog to any specific commercial product, process, or service, is for the information and convenience of the public, and does not constitute endorsement, recommendation, or favoring by the Department of the Interior.



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